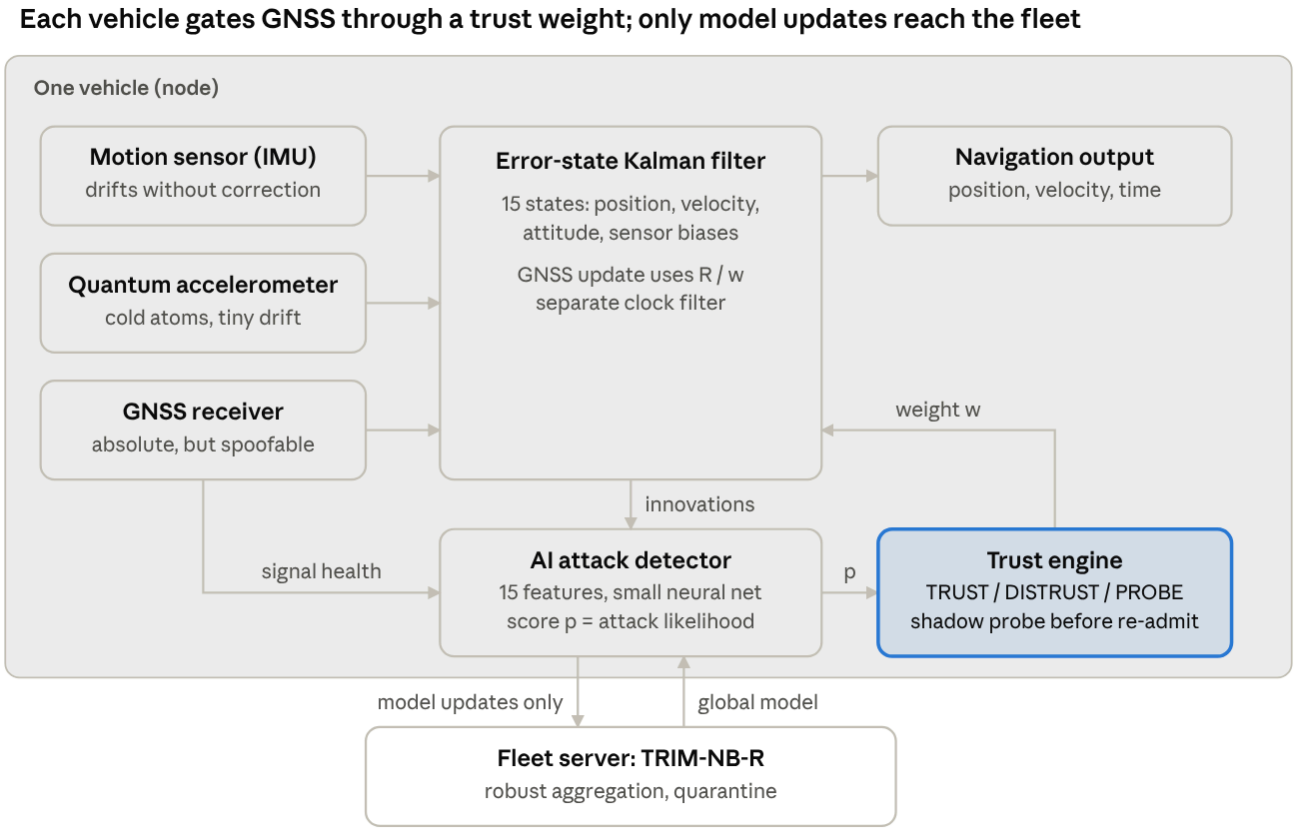
FedQPNT — Technical Overview, Novelty, and Plain Summary

Oct 13, 2026 · @Praty

FedQPNT is a navigation system for vehicles and drones that keeps working when GNSS (GPS) is jammed or spoofed. It combines a classical motion sensor, GNSS and a quantum accelerometer, uses a learned detector shared across a fleet, and decides continuously how much to trust GNSS. A pre-registered test on 5,340 unseen simulation runs showed faster attack detection and clear protection against relayed (meaconing) attacks, while the quantum sensor and fleet learning showed no measurable benefit.

At a glance



FedQPNT architecture · one node and the fleet server

The filter always runs on the IMU and quantum sensor; GNSS enters only through the trust weight w that the detector and trust engine set. Raw data never leaves the vehicle; only detector updates go to the fleet server.

1. Sensors and fusion filter

Each vehicle (a "node") fuses three sensors in one 15-state error-state Kalman filter (ESKF). The same filter is used unchanged by every method and baseline, so any difference in results comes from the trust and learning logic, not from filter differences.

Sensor	What it measures	Strength	Weakness
IMU (accelerometer + gyroscope), two grades: industrial MEMS and tactical	Acceleration and rotation, 100 Hz	Cannot be spoofed	Drifts without correction (MEMS fastest)
GNSS receiver	Position, velocity, clock, plus signal health (C/N0, AGC, satellite count)	Absolute, no drift	Can be jammed or spoofed
Cold-atom interferometer (CAI) accelerometer	Acceleration from atom-wave interference	Bias set by atomic physics, near drift-free	Slow cycle, loses contrast above about 17 mrad/s rotation

Filter states. Position, velocity, attitude, accelerometer bias and gyroscope bias (3 each = 15). A separate two-state clock filter tracks receiver clock bias and drift. Single-vehicle tests use a round-Earth model with Schuler dynamics; fleet tests use a flat-Earth model.

Quantum sensor model. The CAI noise level, its scaling with interrogation time and its bursty fault behaviour are fitted to real laboratory shot data from a published three-axis cold-atom interferometer (d'Armagnac de Castanet et al., 2024), not assumed.

How trust enters the filter. Every GNSS and CAI update uses an inflated noise covariance:

$R_{eff} = R / \max(w, w_{min})$, with $w_{min} = 0.02$

A trust weight w of 1 means full trust; a weight near w_{min} means the measurement barely moves the estimate. A fixed inflation $\kappa_R = 60$ on the receiver-reported GNSS covariance makes the filter's uncertainty consistent on clean data.

Two consistency corrections in the final code. Both make the filter's own uncertainty honest when GNSS is excluded for minutes:

- **Initial tilt uncertainty.** The starting roll and pitch uncertainty is $\max(1\ \text{mrad}, \sigma_{ba} / g)$, because levelling from a biased accelerometer leaves about 10 mrad of tilt error on a MEMS unit.

- **Second-order attitude term.** A horizontal tilt error ψ also produces a vertical acceleration error of about −½ g |ψ|², which the standard linear model ignores. Its exact Gaussian second moment is added to the covariance, with no tuning constant. Without it the filter becomes over-confident during long coasts and can never re-admit GNSS.

2. Detector and trust engine

A small neural network scores every GNSS epoch for attack likelihood, and a trust state machine turns that score, plus physical evidence, into the continuous trust weight w.

Features (15 per epoch). Normalised position and velocity innovations, the RAIM statistic, mean C/N0 and its spread across satellites, C/N0 rate, AGC level, clock-bias and clock-drift jumps, residual RMS, change in satellite count, a CUSUM of the position innovation, an outage flag, the cross-satellite C/N0 correlation over 20 epochs, and the C/N0-versus-elevation slope. The last two capture the single-antenna spoofing signature reported in field data: a spoofer transmits all fake satellites from one antenna, so their signal strengths rise and fall together.

Detector. Features are normalised and expanded with two moving averages and a first difference to 60 inputs. The network is 60 → 16 (tanh) → 2 (sigmoid), with one head for spoofing and one for jamming: 1,010 parameters, about 4 kB. Outputs are Platt-calibrated, and the final score is p = max(p_spoof, p_jam). The final detector (v5) was trained on labelled tuning missions only.

Trust state machine. The GNSS position trust and the GNSS clock trust each follow their own copy of this machine:

State	Entered when	Weight w	Exit
TRUST	Start, or a successful probe	Follows 1 − p; drops fast (τ = 0.5 s), recovers slowly (τ = 10 s)	Alarm held (smoothed p ⩾ 0.6 for 0.5 s) or physical evidence fires
DISTRUST	Alarm or evidence	Pinned at w_min = 0.02 (GNSS effectively excluded; the filter coasts on IMU and CAI)	After T_ex = 60 s, go to PROBE
PROBE	60 s in DISTRUST	Stays at w_min; GNSS is tested but not applied	Pass → TRUST; fail → back to DISTRUST

Shadow probe. During the 10 s PROBE, each GNSS fix is compared with the coasted estimate using the filter's own covariance (a normalised innovation test, bound 25.07 for MEMS and 6.74 for tactical), without letting the fix change the state. GNSS is re-admitted

only if it agrees with the coast and no physical evidence fired. This stops a still-active spoofer from being let back in just because the detector went quiet.

Physical evidence (E_s). Independent of the network: short-baseline position and clock jumps, a cross-satellite C/N0 correlation event, and a C/N0 bump. Each can force DISTRUST. Two short quarantines prevent false triggers: E_s is ignored until the correlation window is full, and the position-jump test is skipped for 2 epochs right after re-admission (the correction pull would otherwise look like a jump).

Guarantee. By construction, trust can cycle at most 223 times per hour, whatever the timing of the evidence.

3. Federated learning across the fleet

The fleet improves one shared detector without any vehicle sending its raw sensor data. Only model updates leave a node.

1. A server starts every node from a pre-trained detector, θ_0.
2. Each node trains locally on its own recent epochs. It needs at least 64 samples before it sends anything.
3. It sends the change to the network weights (a delta), plus its feature-normalisation statistics. Positions, trajectories and raw measurements never leave the node.
4. The server aggregates with **TRIM-NB-R**:
   - clips each update relative to the median update size,
   - takes a coordinate-wise trimmed mean or median,
   - keeps a cosine-similarity reputation per node and quarantines nodes whose updates keep pointing the wrong way,
   - drops updates that are more than 3 rounds stale and down-weights stale ones.
5. Nodes install the new global model, and the next round begins.

What it is meant to resist. Poisoned or faulty nodes: sign-flip, Gaussian-noise, label-flip and "A Little Is Enough" attacks on the learning itself.

Two fixes in the final code. A node that had never installed a global model used to report a base round of −1, so the server treated every one of its updates as too stale and no model was ever installed. It now reports the current round. The pre-trained model's normaliser now comes from the full supervised dataset (20,552 samples) instead of a 1,470-sample subset that caused false alarms on clean fleet nodes.

Important for interpretation. The single-vehicle tests use the frozen trained detector and run no federation rounds. Federated learning is evaluated only in the fleet scenarios and the H2/H4 experiment.

4. Evaluation and results

FedQPNT detects attacks faster and clearly protects against relayed signals, at no cost in normal driving, but it does not improve position accuracy under ordinary spoofing, and the quantum sensor and fleet learning showed no measurable benefit.

Protocol. Everything was pre-registered before the data existed: hypotheses, metrics, statistical tests and pass/fail rules. The design was tuned only on tuning seeds (below 10,000). The code was then frozen (tag core-freeze-6), and 30 never-used test seeds per scenario (10060–10089) were run exactly once at two IMU grades. In total that is 5,340 runs, all completed. The fleet-learning experiment H2/H4 used 10 further unused seeds. Statistics: paired Wilcoxon tests with Holm correction over a fixed family of 8 confirmatory tests, Hodges–Lehmann effect sizes with 95% bootstrap confidence intervals. Two earlier test rounds on separate seeds exposed implementation defects; these were fixed on tuning data only, and the results below come from the fresh, pre-registered third round.

Comparison methods. Undefended (always trusts GNSS); Baseline A (detect-and-exclude: drops GNSS whenever the detector fires); single-node continuous-trust, switch and classical adaptive-filter baselines; and an ablation with the quantum sensor removed.

Scenarios (15 families). Nominal driving, three severities of gradual drift spoofing, a displaced meaconer (an attacker relaying real signals from elsewhere), jamming, jam-then-spoof capture, long GNSS denial with and without the quantum sensor, on/off toggling attacks at 2–60 s periods, noise robustness, recovery after attack, a 4-hour stability run, and fleet scenarios (faulty nodes, packet loss, poisoned learning, simultaneous attacks).

Pre-registered hypotheses.

Test	Question	Result
H1 latency	Detects attack onset faster than Baseline A?	**Yes.** About 0.5 s faster, Holm-adjusted p = 0.0037 (the only significant test)
H1 accuracy	Lower attack-window position error than Baseline A?	No significant difference (p = 0.27)
H2	Federated learning detects a new attack type faster than local learning?	No difference: both arms detect at the first epoch, p = 1
H3	The quantum sensor shortens time to useful accuracy?	No difference at either IMU grade, p = 1

Test	Question	Result
H4	A newly joined node benefits from the fleet model?	No difference on the primary measures, p = 1

Headline numbers (median of 30 seeds, industrial MEMS IMU).

Situation	Measure	FedQPNT	Undefended	Baseline A
Displaced meaconer	Horizontal error during attack	**943 m**	1,342 m	2,154 m
Displaced meaconer	Clock (timing) error	**635 ns**	5,217 ns	1,224 ns
Gradual spoof (medium)	Horizontal error during attack	760 m	**254 m**	623 m
Gradual spoof (medium)	Clock error	743 ns	**10.5 ns**	412 ns
Nominal driving	Horizontal error	about 2.9 m	about 2.9 m	about 2.9 m

What held up.

- No cost in normal driving or under jamming: errors equal to undefended, no false alarms.
- Faster attack detection (H1).
- Displaced meaconing: lower position error than both alternatives on MEMS, and about 8 times lower timing error than undefended.
- Bounded trust cycling: at most 29 cycles per hour observed, against the proven limit of 223.
- 4-hour stability: the last-hour error was 0.79 (MEMS) and 0.84 (tactical) times the first-hour error, so there was no drift.
- The fleet installs shared models with no false alarms on clean nodes.

What did not.

- Under ordinary small-offset spoofing, position error is higher than doing nothing. Excluding GNSS means coasting on the IMU, which drifts further than the spoof pulls the position.
- After an attack, recovery is slower than Baseline A, because of the deliberate 60 s wait before re-testing GNSS.

- The quantum sensor did not measurably reduce coasting error: the ratio without it to with it was 1.006 (MEMS) and 0.986 (tactical).
- Fleet learning showed no detection benefit. The detector could not separate the abrupt-onset attack used for H2/H4 from normal data, so both arms scored identically.
- If real GNSS noise is at least twice what the receiver reports, the system can lock GNSS out on clean data.

5. Novelty: how FedQPNT differs from existing work

No work in our verified literature combines a physical quantum inertial sensor, a federated-learning attack detector and a continuously varying sensor-trust law in one navigation loop; FedQPNT does, and it adds a re-admission test and a pre-registered evaluation that this literature lacks. The claim is limited to the literature we reviewed and verified. No formal patent prior-art search has been done.

Work	Federated learning across vehicles?	Physical quantum sensor?	Continuous trust during an attack?	Tested under attack?
Khan et al. 2025	Yes	No	No (detection only)	Yes, GPS spoofing
Chai et al. 2025	Yes	No	No (classification)	Yes, spoofing and jamming
G²FL, Liu et al. 2026	Yes (robust to bad clients)	No	No	Yes, spoofing
Negru et al. 2024	No (single-vehicle "federated filter")	No	Yes (learned master filter)	No, degraded GNSS only
Pardhasaradhi & Cenkeramaddi 2022	No	No	No (switch to radar)	Yes, GPS spoofing
Hazarika et al. 2025; Chehimi et al. 2023	Yes (quantum computing for learning)	No	No	No
Tariq & Ahanger 2026	No (model ensembling)	No	Partial	Yes, BeiDou spoofing

Work	Federated learning across vehicles?	Physical quantum sensor?	Continuous trust during an attack?	Tested under attack?
FedQPNT	**Yes**	**Yes, cold-atom, calibrated to real data**	**Yes, evidence-bounded**	**Yes, 15 scenario families**

Specific points of difference.

1. **Detector drives a continuous trust weight, not an on/off switch.** Every federated spoofing-detection paper we found turns the detector output into a binary exclude or flag decision. In FedQPNT the detector is one input to a trust law that scales the measurement covariance smoothly.
2. **"Federated" means learning across vehicles.** Several "federated" navigation papers use the classical single-platform federated filter (Carlson, 1988–1990). That is a filter-bank architecture inside one vehicle, not learning across a fleet. FedQPNT does the latter and keeps the two meanings separate.
3. **A real quantum sensor, not quantum computing.** "Quantum federated learning" papers (Hazarika 2025; Chehimi 2023) use quantum computation for the learning step. FedQPNT uses a cold-atom accelerometer as a physical measurement, with noise, scaling and fault behaviour fitted to real laboratory shot data (d'Armagnac de Castanet et al., 2024), including its loss of contrast under rotation.
4. **Shadow-probe re-admission.** Detect-and-exclude methods re-admit GNSS as soon as the detector goes quiet, and adaptive filters re-weight from innovations alone. FedQPNT re-admits only after a separate test of incoming GNSS against the coasted state, without applying it, plus a check that no physical evidence fired. A spoofer that fools the detector is still not let back in.
5. **Separate position trust and clock trust.** Position and timing are trusted independently, so timing can stay protected while position coasts. This is what gives the about 8-times timing improvement under displaced meaconing.
6. **Evidence-bounded exclusion with a proven cycling limit.** Exclusion has a fixed maximum dwell, and trust can switch at most 223 times per hour regardless of the attacker's timing. We did not find an equivalent guarantee in the cited trust-based methods.
7. **A consistent coast covariance that makes re-admission possible.** The second-order attitude term is standard estimation mathematics, but applying it so a consistency-based re-admission test can actually pass after long GNSS exclusions is specific to this design.

8. **Pre-registered, run-once evaluation.** Hypotheses and rules were fixed before data, seeds were used once, and every method shares the identical filter. That level of protocol is uncommon in this literature and makes the results, positive and negative, credible.

What the novelty does not imply. The combination is new, but in this evaluation two of its ingredients, the quantum sensor and federated learning, did not produce a measurable benefit. The demonstrated gains come from the trust law and the re-admission design.

6. Plain explanation (for the review)

The problem. Self-driving cars and drones rely on GPS. GPS signals are weak, so an attacker can jam them (drown them out) or spoof them (broadcast fake signals that make the vehicle think it is somewhere else).

The idea. Give each vehicle a way to notice when GPS is lying and to lean on other sensors instead:

- **A motion sensor (IMU)** that tracks movement on its own but slowly drifts.
- **A quantum accelerometer**, which uses clouds of ultra-cold atoms to measure acceleration with almost no drift.
- **A small AI detector** that watches GPS signal patterns for signs of an attack.
- **A "trust dial"** instead of an on/off switch. When the detector or physical evidence says GPS looks wrong, the dial turns GPS down and the vehicle navigates on its motion sensors. After a minute it quietly checks whether GPS agrees with where the vehicle thinks it is, and only then turns GPS back up.
- **Fleet learning.** Vehicles improve the shared AI detector together by exchanging only model updates, never their location data (federated learning).

How we tested it. We built a full simulator with realistic sensors, the quantum sensor's noise copied from real lab data, and 15 kinds of attack and stress situations. Before running the final test we wrote down exactly what we would measure and how we would judge success. Then we froze the code and ran 5,340 fresh simulations once, comparing against doing nothing and against a simple "drop GPS when the detector fires" method.

What we found.

- It **detects attacks faster** than the simple method, and this was statistically significant.
- It **costs nothing in normal driving** and is safe under jamming.
- Against an attacker who **relays real GPS signals from another place**, it keeps position error lower and timing error about 8 times lower.
- It is **stable**: no drift over 4 hours, and the trust dial does not flicker.

- But against **small, gradual spoofs** it does worse than doing nothing, because coasting on the motion sensor drifts further than the fake signal moves you.
- The **quantum sensor and fleet learning did not give a measurable benefit** in this setup.

One-sentence version. *We built and rigorously tested a GPS-attack-resilient navigation system that uses a continuous trust dial instead of an on/off switch; it detects attacks faster and strongly protects against relayed-signal attacks at no cost in normal operation, and our pre-registered test also shows honestly where excluding GPS does not pay and that the quantum and fleet-learning parts need further work to help.*

Likely questions and short answers.

Question	Answer
Is the quantum sensor real?	Simulated, but its noise and faults are fitted to real laboratory data from a published cold-atom instrument.
Why did it lose under gradual spoofing?	Excluding GPS means coasting on the IMU, which drifts faster than a small spoof displaces the position.
Why no benefit from fleet learning?	The detector could not distinguish the specific abrupt attack used in that test, so both versions scored the same.
Why trust the results?	Pre-registered rules, code frozen, unseen test seeds run once, every method on the identical filter, all code and data on GitHub.
What is new?	The combination of a quantum sensor, fleet-learned detection and a continuous, evidence-checked trust dial with a safe re-admission test.